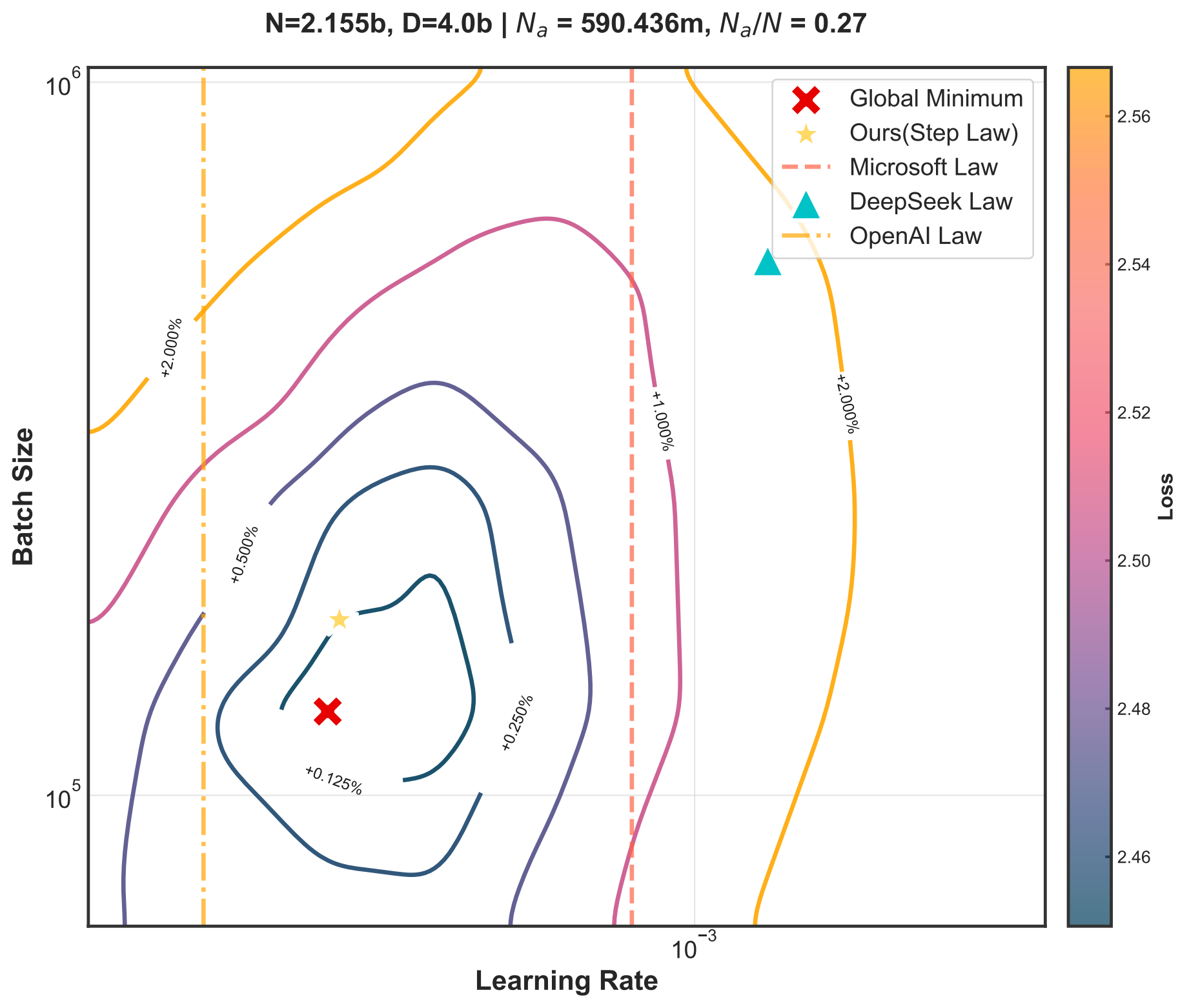
N=2.155b, D=4.0b | N_a = 590.436m, N_a/N = 0.27
Global Minimum
Ours(Step Law)
Microsoft Law
DeepSeek Law
OpenAI Law
+2.000%
+1.000%
+2.000%
+0.500%
+0.250%
+0.125%
10^6
10^5
10^{-3}
Batch Size
Learning Rate
Loss
2.56
2.54
2.52
2.50
2.48
2.46